Software Quality and Project Development

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Shell UNIX

Mandatory to use for servers



Shell UNIX

What is?

What is a Shell?

The shell is a program where users can type commands, where it's possible to create an empty directory with only one line of code.

GUI?

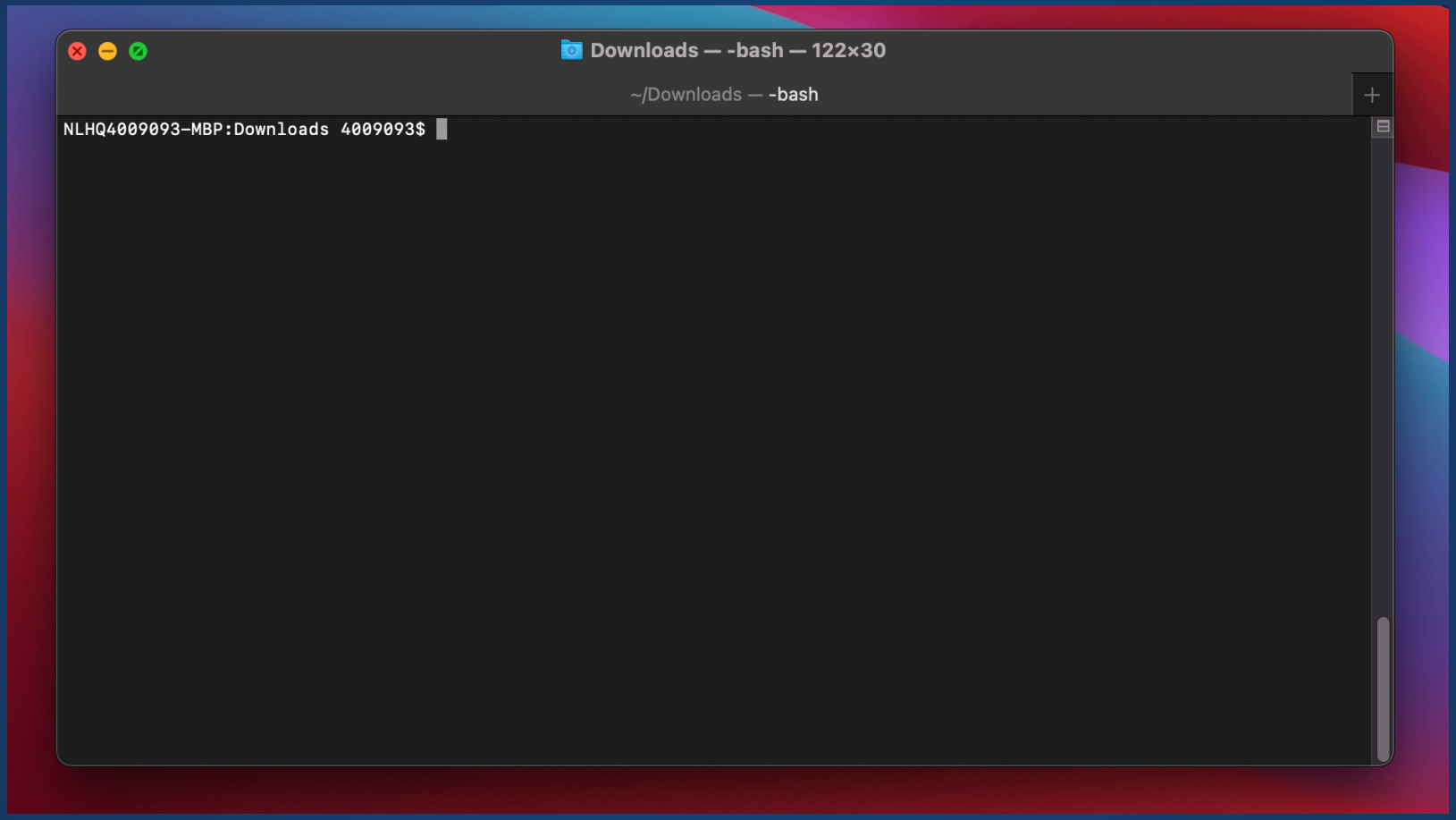
Shell vs GUI

The shell is faster and it allows to learn more than a GUI.

Learning

Learning

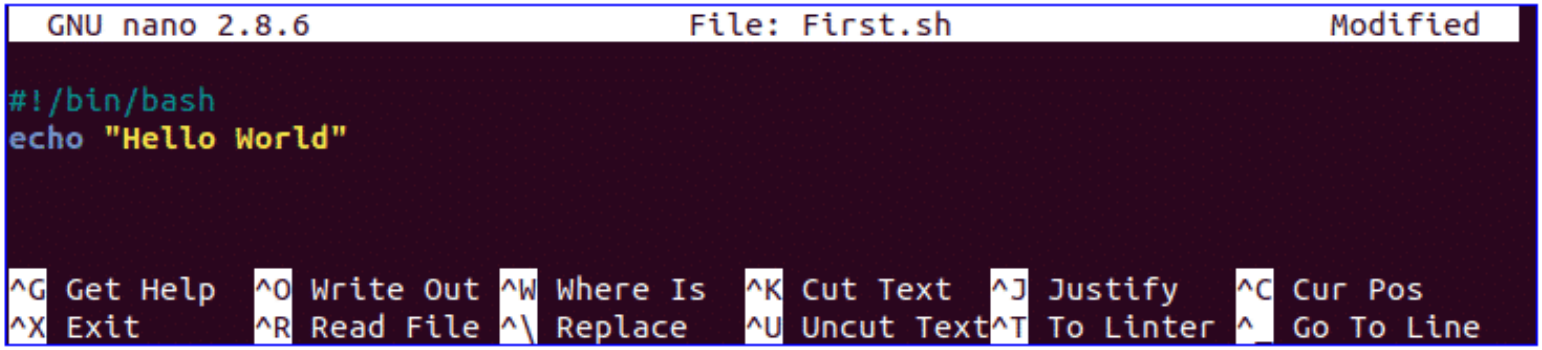
Using a shell allows to be sure about what you are doing without having the mask of a GUI.


Downloads — -bash — 122×30
~/Downloads — -bash
NLHQ4009093-MBP:Downloads 4009093$

```
ubuntu@ubuntu-VirtualBox:~$ echo "Hello World"
Hello World
ubuntu@ubuntu-VirtualBox:~$
```

```
$ nano First.sh
```

```
#!/bin/bash
echo "Hello World"
```




```
$ bash First.sh
```

```
#!/bin/bash

# Add two numeric value
((sum=25+35))

#Print the result
echo $sum
```

```
$ bash comment_example.sh
```

```
ubuntu@ubuntu-VirtualBox:~/code$ bash comment_example.sh
60
ubuntu@ubuntu-VirtualBox:~/code$
```

Useful commands #1

Command	Description
ls	show files and directories
man	read manual
cd	open directories
mv	move (or rename) file
cp	copy file or directory (-r)
rm	remove file or directory (-r)
pwd	(current) path working directory
mkdir	create a directory
nano, vim, ed	terminal text editor

cd

```
$ cd directory_name
```

```
$ cd ..
```

ls

```
$ ls
$ ls path
$ ls -f
$ ls test_*
$ ls *.cpp
$ ls ?.cpp
```

rm

$ rm path/of/file

$ rm -r path/of/directory/

$ rm -rf path/of/directory



The shell does not have a trash bin: once something is deleted, it's really gone.

pwd

$ pwd - return the absolute path of the working directory

/home/helias/ = ~/

Pathnames

.. upper directory

. current directory

absolute path

/home/helias/Desktop/file.png

relative path

home/helias/Desktop/file.png

./home/helias/Desktop/file.png

Desktop/file.png = ./Desktop/file.png

TAB button



superpower: **autocomplete**

Useful commands #2

Command	Description
cat	displays the contents of its inputs
head	displays the first 10 lines of its input
tail	displays the last 10 lines of its input.
command > [file]	redirects a command's output to a file (overwriting any existing content).
command >> [file]	appends a command's output to a file.
[first] | [second]	it is a pipeline: the output of the first command is used as the input to the second.
grep [filter]	grep part of input based on filter

Useful commands #3

Command	Description
Ctrl+C	halts the current command
Ctrl+Z	stops the current command, resume it in the background
Ctrl+R	type to bring up a recent command
Ctrl+L, clear	clean the terminal
!!	repeats the last command
exit	log out of current session
more, less	read a file part by part

Useful commands #4

Command	Description
$@	take all inputs parameters
$1	take the first input parameter
$2	take the second input parameter
wget	"web get", download from URL
curl	transfer a URL, send http requests
zip	it allows to compress directories and/or files
mysql / mysqldump	client mysql

alias

$ man

$ What manual page do you want?

$ woman

$ -bash: woman: command not found

$ alias woman = man

$ woman

$ What manual page do you want?

tmux - cheatsheet

```
$ tmux new -s session-name

$ tmux a -t session-name

$ Cltr+B + D      # detach session
```

Exercises #1

1. make a directory called first
2. change directory to the first folder
3. create a file called person.txt
4. change the name of person.txt to another.txt
5. make a copy of the another.txt file and call it copy.txt
6. remove the copy.txt file
7. make a copy of the first folder and call it second
8. delete the second folder
9. Create a bash script that executes automatically all these tasks waiting 1 second for each task using sleep

Exercises #2

1. What does the man command do? Type in man rm. How do you scroll and get out?
2. Look at the man page for ls. What does the -l flag do? What does the -a flag do?
3. Type the following command to download and save the contents of google.com:

   curl https://www.google.com > google.html
4. Use less to look at the contents of google.html.
5. Look at the man page for less. Read the section on -p (/pattern). Search for the text hplogo in the google.html file.
6. What is the shortcut to clear the terminal?
7. What is an absolute path?
8. What is an relative path?
9. What do the r and f flags do with the rm command?

alias exercises

$ zipthis DIRECTORY

$ notify MESSAGE

$ echo "hello world" | espeak -v it

Challenge: extend notify-me

Extend the bash script notify.sh into other bash scripts that allow to send single file or directory (zipped), using:

$ *tg-send path/of/the/file/or/directory*

Bonus feature:

- recognize if the file is more than 50 MB, in that case the Telegram API will block you, so it is better to warn the user about it

- use telegram client (not bot) API to overcome the 50 MB limit and get 2 GB as limit.

References:

- https://github.com/Helias/Notify-me

- https://github.com/azerothcore/telegram-automated-db-backup